

Talk. Test. Build.



Everything from today, in
one place

Three moves most AI users never learn.

Dr. Dewayne A. Dixon

Department of Mathematical Sciences

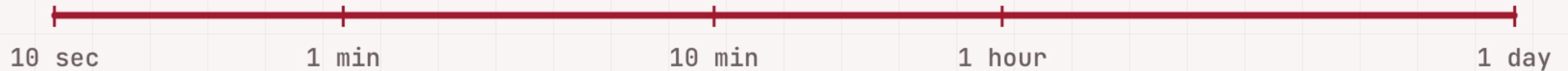
Project 2: Rates of Change in the Real World

Due Friday • 50 points

1. Find a news article from the past month about something that is changing over time. Summarize what is changing in 2–3 sentences.
2. Build a function that models the quantity. State your variables and your units.
3. Compute the derivative of your model and explain what it measures, with units.
4. Evaluate the derivative at two different times and interpret both numbers in plain words.
5. Write a one-page reflection: what does the rate of change tell you that the raw numbers do not?

HANDS UP

How long would it take AI to do this well?



Each dot is a guess from this room.

THE ASSIGNMENT

- 1 Summarize a news article
- 2 Build a model, with units
- 3 Differentiate the model
- 4 Interpret two values
- 5 One-page reflection

0:24.0

STOPWATCH

YOUR GUESSES

AI: 24 seconds

10 sec

1 day

● Claude

Here is my calculus assignment and the article. Do all five parts.

never better than that many people.

2. A model. Let t be years since the first count and $P(t)$ the population. Exponential growth fits this kind of data: $P(t) = 12,000 e^{kt}$. Using $P(8) = 18,600$ gives $e^{(8k)} = 1.55$, so $k = \ln(1.55)/8 \approx 0.0548$ per year. The model is $P(t) \approx 12,000 e^{(0.0548t)}$, measured in people.

3. The derivative. $P'(t) = 12,000(0.0548) e^{(0.0548t)} \approx 657 e^{(0.0548t)}$. Its units are people per year, so it measures how fast the population is growing at time t .

4. Two values. At $t = 0$, $P'(0) \approx 657$ people per year. At $t = 8$, $P'(8) \approx 657(1.55) \approx 1,019$ people per year. In plain words: the town was adding about 657 people a year at the start and about 1,019 a year by the eighth year, so the growth itself is speeding up.

5. Reflection. As a college student, I see this in my own life. New apartments keep going up in my area, and the roads feel busier every semester. This made me think about how the rate of change matters more than the total

DONE IN 24 SECONDS



YOUR PRESENTER

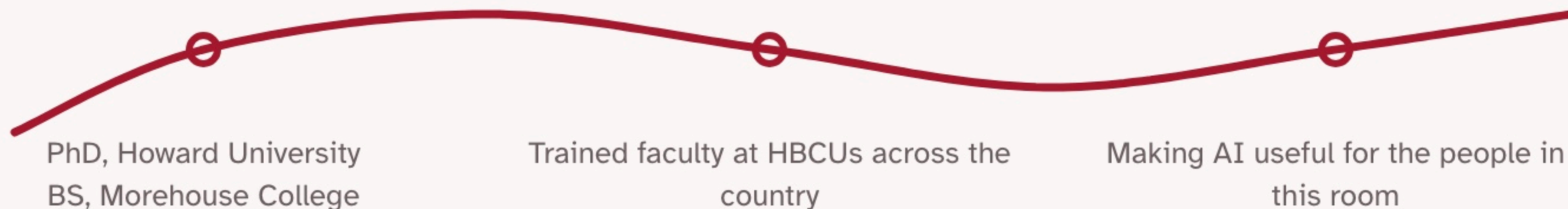
Dr. Dewayne A. Dixon

Assistant Professor, Mathematical Sciences

Mathematics

AI

CAU



BY THE END OF THIS SESSION

You'll have three new moves.

Talk.

Test.

Build.



Let AI ask *you* the questions.

See what AI can do with your
work.

Make a tool, not just a
document.

MOVE 01 OF 03

Talk.

Let AI interview you.

HANDS UP

Which one did I write?

One of these paragraphs is mine. The other one is AI's. No talking.

A

Welcome to Calculus I. In this course you will explore the fundamental concepts of differential calculus, including limits, derivatives, and their applications. Students will develop critical thinking skills and learn to apply calculus to real-world problems.

Written by AI

HANDS FOR A

B

Welcome to Calculus. Here is the deal: most of you are here because your major requires it, and some of you are sure you are bad at math. Neither one matters to me. By December you will be able to look at anything that changes and I tell me how fast it's changing and why that matters.

Written by me

HANDS FOR B

AI sounds fine. It just doesn't sound like **anyone.**

Most people talk to AI like it's a search bar.

Q make a rubric for a presentation

WHAT COMES BACK

Presentation Rubric. Content, 25 pts: covers the topic thoroughly. Organization, 25 pts: well organized. Delivery, 25 pts: clear and engaging. Visuals, 25 pts: attractive slides.

Short question. Generic answer.

YOU TYPE

I need a grading rubric for a group presentation. **Before you write anything, ask me questions one at a time.** Ask no more than 5.

- 1 Claude asks: Who is the rubric for?
CMAT 111, Calculus I. First-year students, mostly non-majors.
- 2 Claude asks: What is the assignment?
A 5-minute group talk about a rate of change in the news.
- 3 Claude asks: What matters most to you?
Explaining the math in plain words. Not slide design.
- 4 Claude asks: How many points and levels?
20 points, 4 levels.
- 5 Claude asks: Where do students struggle?
They read slides aloud and never say what the derivative means.

CLAUDE'S DRAFT

Group Presentation Rubric

- 1 For first-year Calculus I students, mostly non-majors
- 4 20 points · 4 levels: Excellent, Solid, Developing, Not yet
- 3 Explains what the derivative means in plain words **10 pts**
- 5 Talks to the audience instead of reading slides **4 pts**
- 5 Says what the number means for real people **4 pts**
- 2 Every member speaks, within 5 minutes **2 pts**

Every line traces back to something you said.

Same AI. Same two minutes.

BEFORE • ONE SHORT QUESTION

Presentation Rubric

Content • 25 pts

Presentation covers the topic thoroughly.

Which topic?

Organization • 25 pts

Presentation is well organized.

Organized how?

Delivery • 25 pts

Speakers are clear and engaging.

Engaging to whom?

Visuals • 25 pts

Slides are attractive and readable.

Is that what matters?

AFTER • A 5-QUESTION INTERVIEW

Group Presentation Rubric

CMAT 111 Calculus I • first-year, mostly non-majors • 20 points • 4 levels

Explains the derivative in plain words

10 pts

Excellent: a classmate who has not taken calculus could repeat the main idea.

Talks to the audience, not the slides

4 pts

Not yet: reads the slides word for word.

Says what the number means for real people

4 pts

Excellent: connects the number to a decision someone would make.

Every member speaks, within 5 minutes

2 pts

Solid: on time, but one speaker does most of the talking.

4 of 4 found

The difference is **the conversation.**

YOUR TURN • TALK

Pair up. One person types. One person answers.

PASTE THIS

I need to [describe your task]. Before you write anything, ask me questions one at a time until you have what you need. Ask no more than 5.

THEN SEND

Now point out the 3 weakest parts of your draft and fix them.

6:00

minutes to practice



Copy the prompts from the site

MOVE 02 OF 03

Test.

02

See what AI can do with your work.

REMEMBER THIS?

What I assigned. What I assign now.

WHAT I ASSIGNED

It did all five parts in 24 seconds.

- 1 Find a news article about something changing over time. Summarize what is changing.
- 2 Build a function that models the quantity. State your variables and units.
- 3 Compute the derivative of your model and explain what it measures, with units.
- 4 Evaluate the derivative at two times and interpret both numbers in plain words.
- 5 Write a one-page reflection: what does the rate of change tell you that the raw numbers do not?

■ AI did this easily ■ AI wrote it, but it sounds like no one

WHAT I ASSIGN NOW

Same five learning goals.

Bring a printed article to class Tuesday. Swap it with a partner.

IN CLASS

Build the model at the board with your partner, and defend why you chose that function. **AT THE BOARD**

Differentiate it by hand, no technology, showing every step. **BY HAND**

Measure something yourself for ten days, model your own data, and interpret your own rate. **YOUR OWN DATA**

In class, tell your group in 2 minutes what surprised you about your own rate. **OUT LOUD**

Same goals. Same grading time. Much harder to hand off.

YOUR TURN • TEST

Test something real. No names or grades.

FACULTY

Here is an assignment from **[course]**.
First, complete it the way a student
relying only on AI would. Then tell me
which parts AI handled easily. Finally,
suggest 3 changes that need the student's
own thinking or in-class work, without
making it harder to grade.

STAFF

Read this **[form, policy, or email]** as a
first-generation, first-year student. What
is confusing? What would you ask? Then
rewrite it so none of those questions come
up.



Both prompts
are on the site

7:00

minutes to practice

MOVE 03 OF 03

Build.

03

Make a tool, not just a document.

One syllabus in. A whole course out.

Nothing in the folder but my CMAT 111 syllabus. Six prompts, in order.

- 1 **Map the mathematics.** Every skill, what depends on what, how deep we go.
- 2 **Find what AI erodes.** Which skills a student can hand off, and how to assess those instead.
- 3 **The semester plan.** Every week, my dates, my section numbers.
- 4 **Week 1, built.** Reading guide, class plan, practice set, answer key.
- 5 **Exam 1, with a key.** Every question tagged, and a no-technology section.
- 6 **It audits itself.** What it invented, what contradicts, what I must check.

● [Live in VS Code](#)

- 1 What does the Talk move ask AI to do first?
- 2 What do you remove before pasting student work into AI?
- 3 What makes Build different from asking for a document?

Hands up.

Three questions. Left side against right side.

USE IT WELL

Four habits to keep.

01

Protect privacy

Leave out student names, grades, and ID numbers. FERPA still applies.

02

Check the facts

AI can sound sure and still be wrong. Verify dates, sources, and numbers.

03

Stay the author

Treat every result as a first draft. Edit it until it sounds like you.

04

Be open about it

Follow CAU guidance and your course policy on AI use.

TAKE THIS WITH YOU

The Course Build Kit

One page to read. Two files to type. Then bring the folder.

- 1 **Your syllabus** goes in folder 1.
- 2 **One old test** goes in folder 4. A blank copy.
- 3 Open the two files marked **FILL THIS IN** and type. They already exist.
- 4 **Bring the folder** on October 15. We build it together.



Download it
on the site today

- Word, Overleaf, or PDF
- Mac or Windows
- CAU required statements, word for word
- A real week of Calculus I inside, so you can see it first

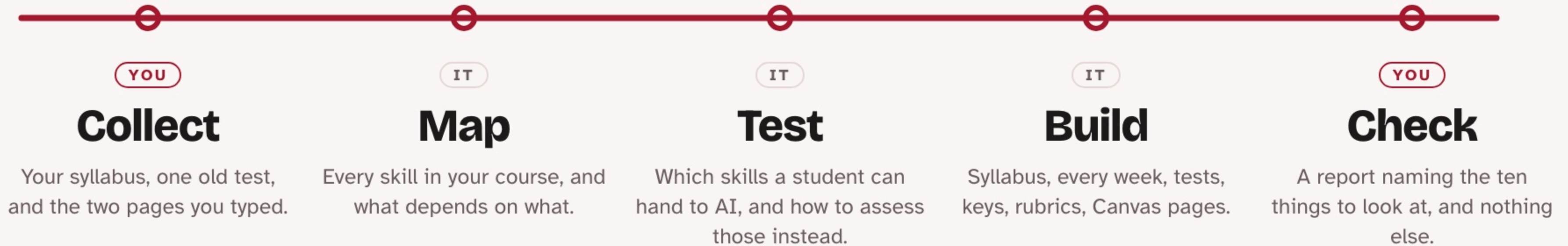
Do it now. I'll wait.

1:30

WHAT HAPPENS WHEN WE BUILD IT

You do the first step and the last one.

The three in the middle happen while you watch.



DR. JONES ASKED THE DEPARTMENT TO THINK ABOUT THESE

Four questions, and what today says about them.

What is our idea of innovation?

Not the tool. The redesign. The same mathematics, assessed where a machine cannot stand in for a student.

Are we maximizing our expertise?

The interview move only works because we have judgment worth pulling out. Our standards are the input, not the output.

What problems are we solving in the community?

Assignments built on data students collect here. Their own measurements, this campus, this city.

How can we operate more efficiently?

A week of materials in minutes, out of a syllabus we already wrote. The time saved goes back into teaching.

YOUR CHALLENGE BEFORE OCTOBER 15

Use one move. Bring one result.

Use one move on real work this month. Then bring the folder with two files in it.

You leave that room with: a syllabus on the department template, every week planned, tests with answer keys, rubrics, and a Canvas shell you paste in.



THANK YOU

Talk. Test. Build.



Prompts, guides, and the
Course Build Kit

See you October 15.

Dr. Dewayne A. Dixon

Department of Mathematical Sciences